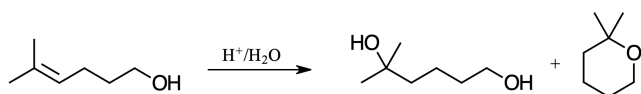


Mechanism Questions #0

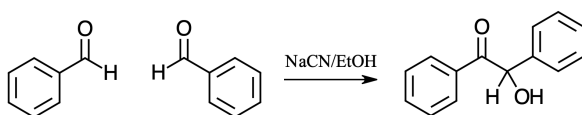
Barry Linkletter

Be sure to carefully consider your proposed electron movement. Are your arrows correct? Are your intermediates reasonable?

1. When using isotopically labeled water, the hydrolysis of epoxide gives different products depending on if the reaction is performed in acidic or basic solution (See figure 1). Can you explain the observation by proposing a mechanism for each reaction? How would you exploit this observation in organic synthesis?
2. Explain the mechanisms that give rise to each of the products below. What reaction conditions would you choose to maximize the yield of each?



3. Bromination of acetone in water is fast but the rate is independent of bromine concentration. (See figure 3). Can you explain the observation by proposing a mechanism for the reaction?
4. The benzoin condensation seems to be impossible. And it is, unless catalyzed by cyanide ion. Propose a mechanism. Be able to argue for the stability of your intermediates.



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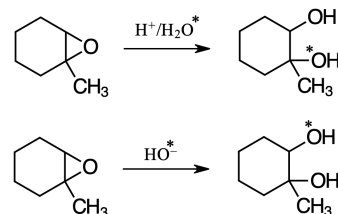


Figure 1: Acid and base catalyzed hydrolysis of an epoxide.↑

Figure 2: Acid catalyzed addition to an alkene.

←

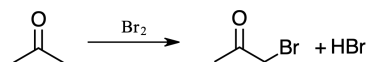


Figure 3: α -bromination of acetone.↑

Figure 4: The benzoin condensation.

←